

SAMUEL BUCKNER

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EDUCATION

Ph.D., *Aeronautics & Astronautics* | University of Washington

Sept. 2021 - Aug. 2026

GPA: 3.73/4.0

- Advisor: Behçet Açıkmеше (Autonomous Controls Laboratory)
- Fellowships: NSF Graduate Research Fellowships Program (GRFP), Achievement Rewards for College Scientists (ARCS)

B.S., *Aerospace Engineering* | University of Florida

Awarded Dec. 2020 (*Summa Cum Laude*)

GPA: 3.80/4.0

SKILLS

General: GN&C, Convex Optimization, Robotics, Dynamics & Modeling, Simulation, Machine Learning
Programming: Python, Julia, C/C++, MATLAB/Simulink, Git, Unix/Linux, $\text{\LaTeX}$
Frameworks & Libraries JAX, ROS, MuJoCo, CVXPY/JuMP, PyTorch
Software/Tools: Unreal Engine/AirSim, Docker, Systems Tool Kit (STK), SolidWorks
Hardware: Machining, 3D Printing, Prototyping, Soldering

WORK EXPERIENCE

Robotics Research Intern | Mitsubishi Electric Research Laboratories

May 2025 - September 2025

Cambridge, MA

- Developed a structure-exploiting algorithm to solve contact-implicit trajectory optimization (CITO) problems using sequential convex programming (SCP) with homotopy logic to handle mixed-integer constraints (contributed to Buckner, 2026).
- Led development of a Python toolbox to solve CITO problems with JAX-based GPU acceleration and customized convex subproblem parsing (awaiting patent approval); demonstrated up to $19\times$ solve-time improvements over existing state-of-the-art SCP solvers.
- Validated solver outputs in MuJoCo environment; observed up to 10% optimality improvement relative to MuJoCo-MPC benchmark.

Graduate Pathways | NASA Johnson Space Center

June 2022 - May 2025

Houston, TX

EG5: Stochastic Trajectory Optimization & Navigation Flight Software Development

- Developed a new approach to guidance/navigation co-design using SCP-based stochastic trajectory optimization with a simultaneous localization and mapping (SLAM) filtering model (ACT-SLAM) submitted under NTR MSC-27989-1.
- Validated the approximative model on lunar landing challenge problems up to 98.3% accuracy.
- Integrated navigation filtering algorithms (MEKF) into a cFS (C++) based flight software for the HyperSTEP platform.
- Supported development of a baseline GN&C framework for the ROS-based STAR robotic platform with hardware validation.

D316 (LaRC): eVTOL Path Planning

- Developed a model predictive control (MPC) framework with SCP for long-horizon reference tracking of an eVTOL vehicle.
- Customized implementation for CFD-based switched dynamics models.
- Validated the approach on generalized urban air mobility (GUAM) dataset scenarios with robust long-duration tracking performance.

EG5: Hazard-Aware Lunar Trajectory Optimization

- Developed new techniques for convexification of sensor-pointing constraints based on robust optimal control; achieved 99% IPM convergence success rate relative to previous polytopic modeling technique (57% success rate).
- Integrated resulting algorithm into Astrobotic's Griffin lunar lander framework; chosen as candidate algorithm for Griffin 2+ missions.
- Published theory and results at SciTech 2024 forum (Buckner, 2024; Shaffer, 2024).

EG5: Hazard-Aware Lunar Trajectory Optimization

- Consulted with Astrobotic to develop multiple new convex-optimization-based guidance algorithms to model state triggered constraints (STCs) for real-time hazard-aware autonomous precision landing in a lunar environment.

GN&C Intern | Blue Origin

Jan. 2021 - Sep. 2021

Kent, WA

Rotation #2: New Glenn Navigation Team

- Developed Simulink/C++ (flight code) implementations of VAR forecasting for autonomous vehicle state prediction & estimation.
- Performed 3σ MC performance analyses, implemented uplink data compression methods and conducted noise sensitivity studies.

Rotation #1: New Glenn Fluid Controls Team

- Developed Simulink unit testing suites and toolkits for fluid control applications, performed off-nominal and uncertainty controller response studies and improved plant dynamics modeling validated against CFD results.

[Undergraduate Pathways](#) | NASA Johnson Space Center

📅 Aug. 2018 - Aug. 2020

📍 Houston, TX

EG6: GN&C Software Development for the ROSIE Robotic Platform

- Co-developed 6-DOF platform robot C++ codebase with PID Control, trajectory sequencing, and OptiTrack+LiDAR sensor integration.

CM45: COMPASS Flight Simulator Development

- Developed Python/C++ lunar ascent simulator incorporating linear tangent guidance and high-fidelity sensor models.

EG4: Integrated GN&C Analysis for the Commercial Crew Program

- Conducted trade studies on algorithms for optimal control allocation and parachute deployment used for the SpaceX Crew Dragon.

[Fit2Fly Research Associate](#) | NASA Langley Research Center

📅 May 2019 - Aug. 2019

📍 Hampton, VA

- Developed multi-UAV SWIL simulator and PyTorch-based RF tracking (98% accuracy) for netted quadrotor flight demonstration.

RESEARCH EXPERIENCE

[NSF Fellow](#) | Autonomous Controls Laboratory (ACL)

📅 Aug. 2021 - Present

📍 Seattle, WA

- Extended on ACT-SLAM formulation and showed up to 36.3% improvements relative to benchmarks; received best paper award in the SciTech GN&C student paper competition (Buckner, 2026).
- Developed conic intersection theory to provide convergence guarantees for robust pointing algorithms (Buckner, 2024; Shaffer, 2024).
- Developed a graph-based trajectory optimization framework (Graph-SCvx) for perception-aware contingency planning; achieved $\geq 95\%$ success rates for quadrotor landing in complex terrain environments where other methods fail (Buckner, 2026).
- Developed general-purpose Julia package for Graph-SCvx and related algorithms (DDT0Too1Box.jl) along with two separate ROS packages to facilitate high-fidelity simulation and quadrotor flight testing.
- Other research: CVaR-based stochastic trajectory optimization, learning-based methods for multi-object tracking

[Undergraduate Research Assistant](#) | Advanced Autonomous Multiple Spacecraft (ADAMUS) Laboratory

📅 Apr. 2018 - Nov. 2020

📍 Gainesville, FL

- Developed a CubeSat flight software involving a ROS (Robot Operating System) framework for the Drag-Deorbit Device (D3) mission.

SELECTED PUBLICATIONS

- [1] **Buckner, Samuel C**, John M Carson, Breanna J Johnson, and Behcet Acikmese (2026). "Active Continuous-Time Simultaneous Localization & Mapping for Powered Descent Guidance Maneuvers". In: AIAA SCITECH 2026 Forum.
- [2] **Buckner, Samuel C** and Purnanand Elango (2026). "GPU-Accelerated Continuous-Time Successive Convexification for Contact-Implicit Legged Locomotion". In: IEEE. 2026 IEEE International Conference on Robotics and Automation (ICRA).
- [3] **Buckner, Samuel C**, Chris Hayner, and Behcet Acikmese (2026). "Graph-SCvx: Graph Successive Convexification For Deferred-Decision Trajectory Optimization". Submitted to *IEEE Transactions on Robotics*.
- [4] **Buckner, Samuel C**, Joshua Shaffer, John M Carson, Breanna J Johnson, Ronald R Sostaric, and Behcet Acikmese (2024). "Constrained Visibility Guidance for 6-DOF Powered Descent Maneuvers with Terrain Scanning using Sequential Convex Programming". In: AIAA SCITECH 2024 Forum, p. 1759.
- [5] Echigo, Kazuya, Oliver Sheridan, **Samuel Buckner**, and Behçet Açıkmeşe (2024). "Dispersion Sensitive Optimal Control: A Conditional Value-at-Risk-Based Tail Flattening Approach via Sequential Convex Programming". In: *IEEE Transactions on Control Systems Technology*.
- [6] Shaffer, Joshua, Chris Owens, Theresa Klein, Andrew D Horchler, **Samuel C Buckner**, Breanna J Johnson, John M Carson, and Behcet Acikmese (2024). "Implementation and Testing of Convex Optimization-based Guidance for Hazard Detection and Avoidance on a Lunar Lander". In: AIAA SciTech 2024 forum, p. 1584.
- [7] Hayner, Christopher R, **Samuel C Buckner**, Daniel Broyles, Evelyn Madewell, Karen Leung, and Behçet Açıkmeşe (2023). "HALO: Hazard-Aware Landing Optimization for Autonomous Systems". In: IEEE. 2023 IEEE International Conference on Robotics and Automation (ICRA), pp. 3261–3267.
- [8] **Buckner, Samuel**, Carlos Carrasquillo, Marcus Elosegui, and Riccardo Bevilacqua (2020). "A Novel Approach to CubeSat Flight Software Development Using Robot Operating System (ROS)". in: 34th Annual AIAA/USU Small Satellite Conference.